



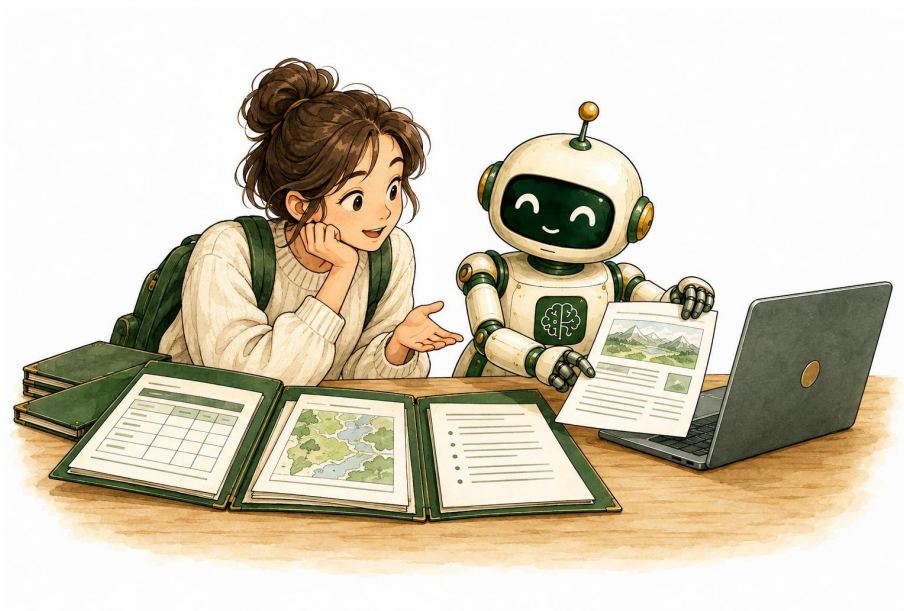
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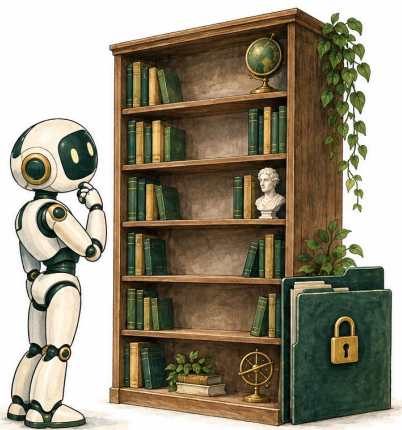
Context Engineering

Retrieval-Augmented Generation

COMP7607A · LECTURE 4



Language model limitations



Knowledge gaps

Domain facts and updates

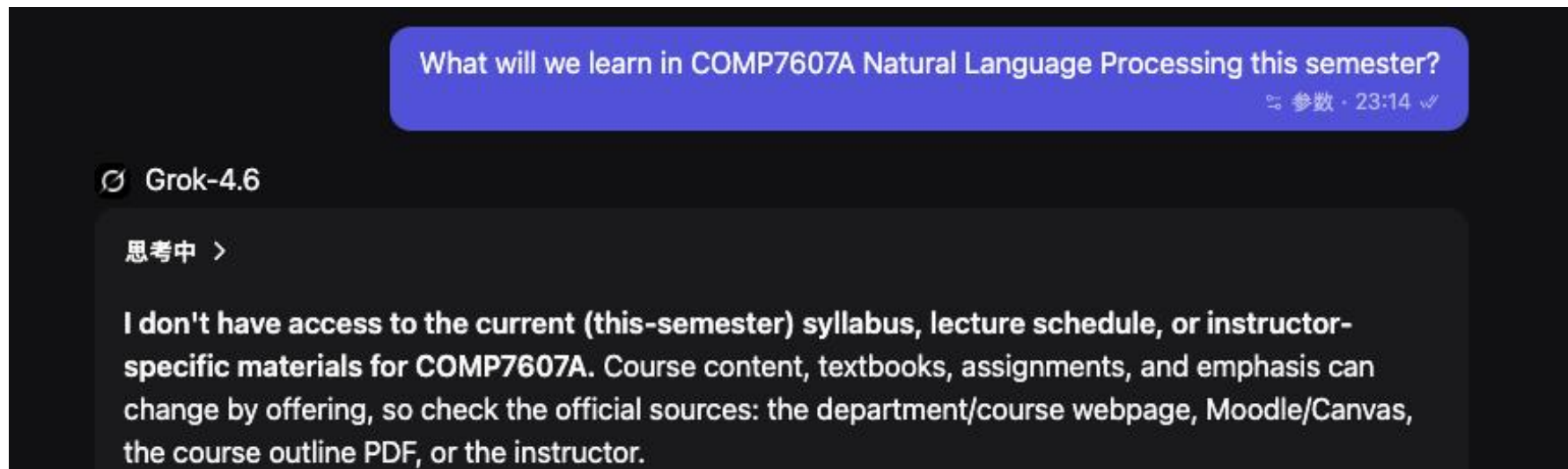


Finite context window

Only so much fits in one input

Limitations: course-specific information

Does the model know our current syllabus?



What will we learn in COMP7607A Natural Language Processing this semester?

参数 · 23:14 ✓

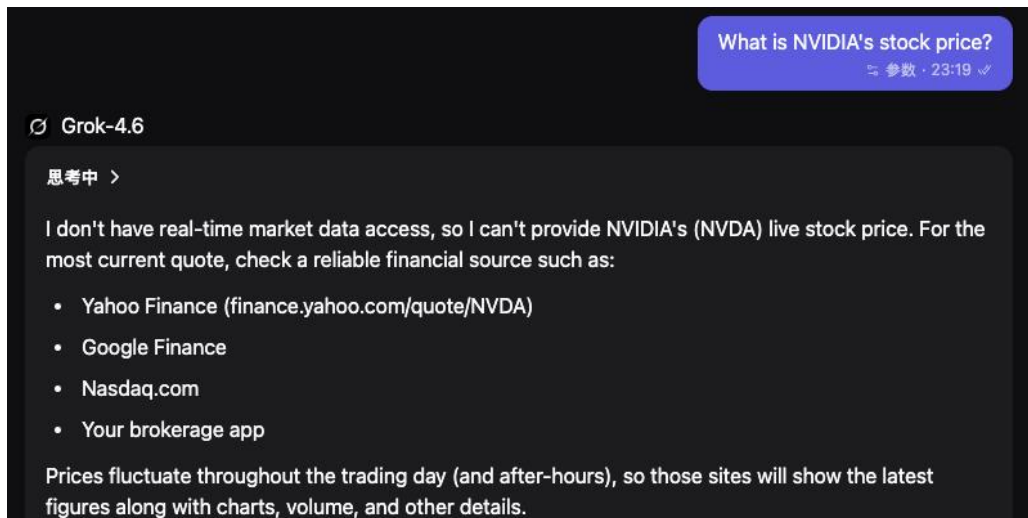
Grok-4.6

思考中 >

I don't have access to the current (this-semester) syllabus, lecture schedule, or instructor-specific materials for COMP7607A. Course content, textbooks, assignments, and emphasis can change by offering, so check the official sources: the department/course webpage, Moodle/Canvas, the course outline PDF, or the instructor.

The answer needs our current syllabus.

Limitations: fresh information



A current question needs a current source.

Why not update the model for every new fact?



Knowledge expires

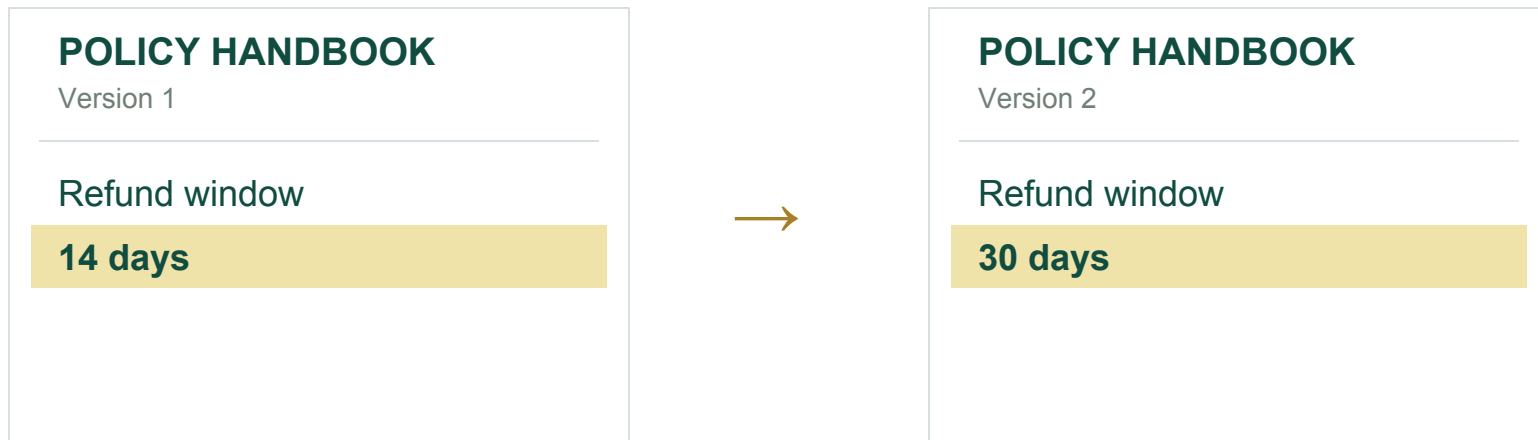


Training costs



Privacy & access

Knowledge can change after training



The answer needs the current source.

Private knowledge needs access control

CONFIDENTIAL FILE

Only the project team

Client records

Restricted access

Authorised user

Retrieve permitted records

Other users

No access to these records

People look up changing facts

“What is NVIDIA’s stock price today?”



Look it up when the question arrives.



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CONTEXT ENGINEERING · LECTURE 4

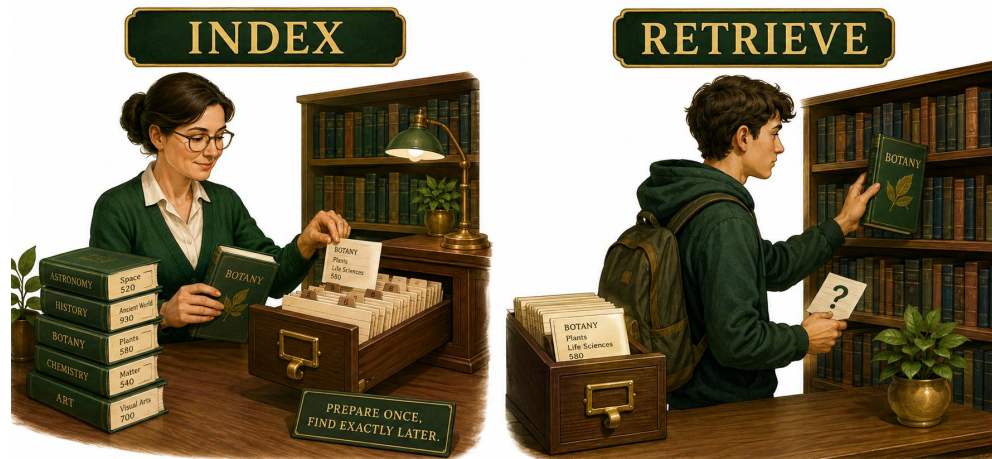
02

Standard framework

SO FAR

We have a collection of course materials.

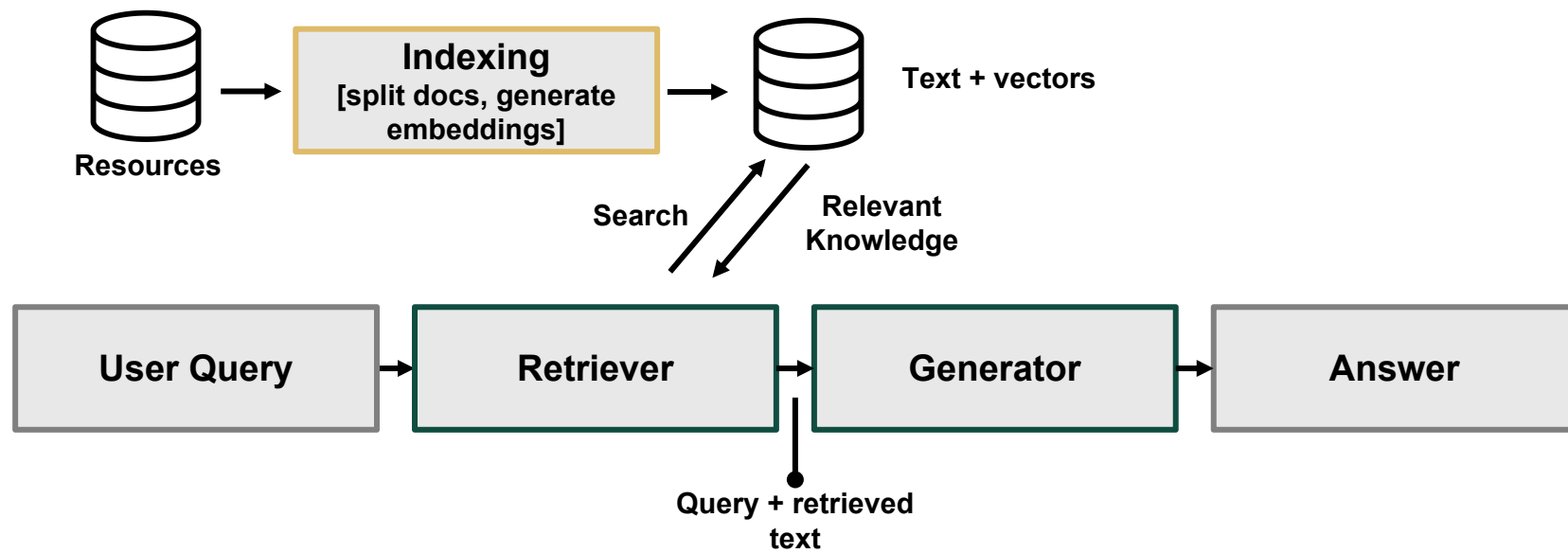
PART 2 OF 6



OUR NEXT QUESTION

How does a question become an
evidence-based answer?

RAG has three core modules





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CONTEXT ENGINEERING · LECTURE 4

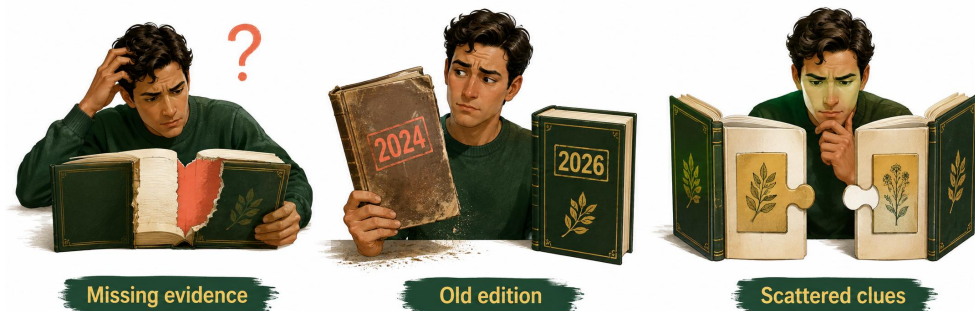
03

Naive RAG

SO FAR

We know the three core modules.

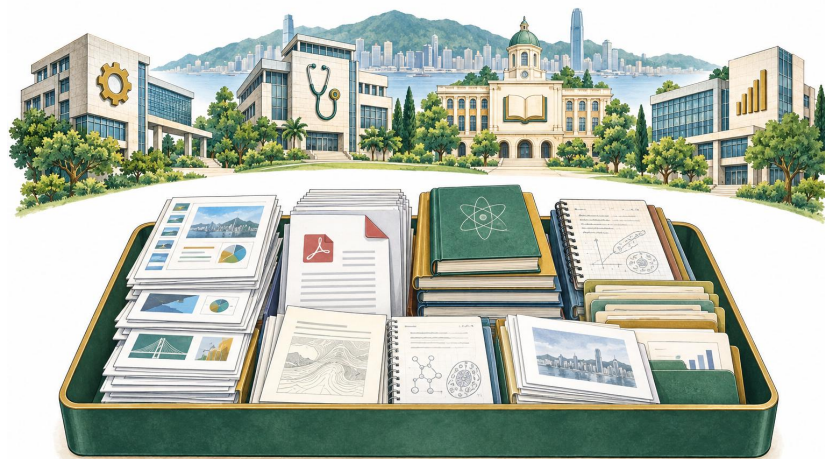
PART 3 OF 6



OUR NEXT QUESTION

When is COMP7607A's final project due?

All HKU course materials



The answer lives inside one document

QUERY When is COMP7607A's final project due?

1 **Collect**

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

All HKU course materials

One question

One relevant document

COMP7607A

2026–27

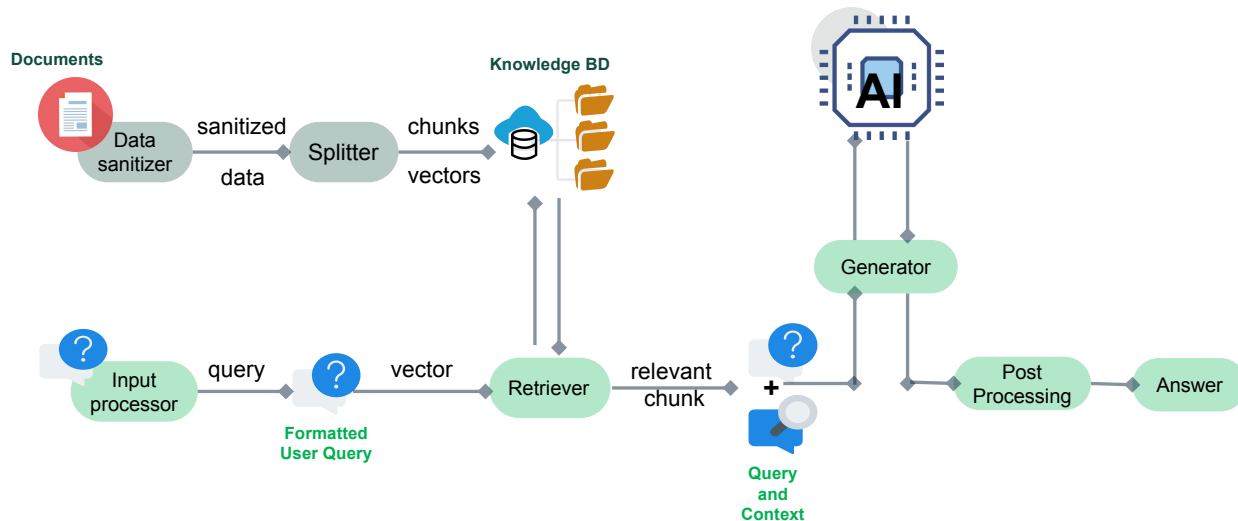
Final project requirements

Deadline: 27 Nov, 17:00 HKT

Submit a report and code

The working pipeline

QUERY When is COMP7607A's final project due?



Our first RAG implementation

QUERY When is COMP7607A's final project due?

```
1 files = collect(all_authorized_course_files)
2 docs = deduplicate(clean(extract_text(files)))
3 chunks = split(docs, keep_metadata=True)
4 index.add(encode(chunks), text=chunks)
5 hits = index.search(encode(question))
6 evidence = rank_and_select(hits)
7 answer = llm(question, evidence)
```

Clean the files, keep the content

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer



Noise out. Course content stays.

```
docs = clean(extract_text(all_course_files))
```

Cleaning in code

QUERY **When is COMP7607A's final project due?**

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

```
function sanitize(paths):  
    docs = []  
    for p in paths:  
        t = extract_text(p)      # text from pdf/doc/html  
        t = clean_text(t)        # remove noise/whitespace  
        t = mask_pii(t)          # redact PII  
        docs.append(t)  
    return docs
```

Extract text. Remove noise. Mask sensitive fields when appropriate.

One copy of each document

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

COMP7607A

Current version

Project brief

Copy A

COMP7607A

Current version

Project brief

Copy B



COMP7607A

Current version

Project brief

One canonical copy

docs = deduplicate(docs, within_same_version=True)

Split every course document

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer



COMP7607A

2026–27

Project deadline

27 Nov, 17:00 HKT

LINGUISTICS

2026–27

Project deadline

20 Nov, 17:00 HKT

All courses

`chunks = split_by_heading(docs, keep_metadata=True)`

Chunking in code

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

```
function split(doc, size=400, overlap=80):  
    chunks = []  
    buf = []  
    for block in doc.blocks:  
        buf.append(block)  
        if tokens(buf) >= size:  
            chunks.append(buf)  
            buf = tail(buf, overlap)  
    return chunks
```

Before return, keep the tail: if buf: chunks.append(buf)

The date needs its heading

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

COMP7607A

2026–27

Final project

27 Nov, 17:00 HKT

“27 Nov, 17:00”

Which course?
Which assignment?

chunk = heading + relevant_paragraph

Every chunk keeps its address

QUERY **When is COMP7607A's final project due?**

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

COMP7607A

2026–27

Final project

27 Nov, 17:00 HKT

course: COMP7607A

year: 2026–27

file: project_brief.pdf

page: 3

Different words, the same question

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

STUDENT ASKS

**“When must I
hand it in?”**

DOCUMENT SAYS

**“Submission
deadline”**

Embeddings turn meaning into numbers

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

Question

“When is it due?”

[0.8, 0.6, ...]

Course text

“Submission deadline”

[0.7, 0.5, ...]

vectors = encoder.encode(chunks)

Same retrieval
encoder

The index stores text and vectors together

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer



```
function upsert(chunks, embed, db):  
    for c in chunks:  
        vec = embed(c.text)           # vector embedding  
        db.upsert(id=hash(c), vec=vec, text=c.text)
```

vectors + original text + source metadata

Similar meanings point in similar directions

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

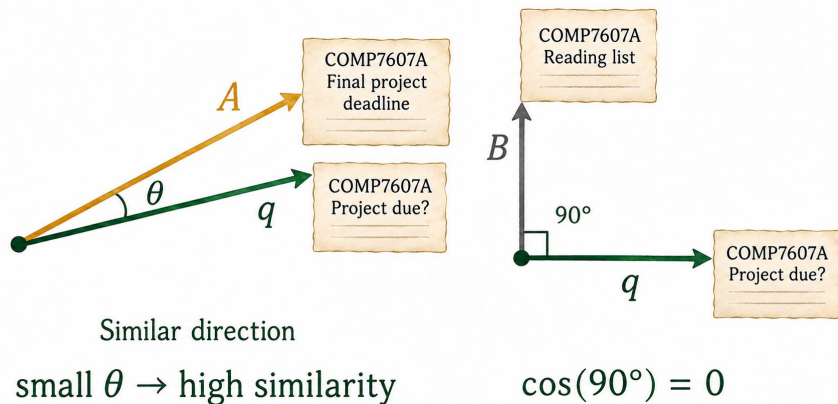
3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer



A: project deadline. B: reading list. Which answers our query?

Cosine similarity compares directions

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

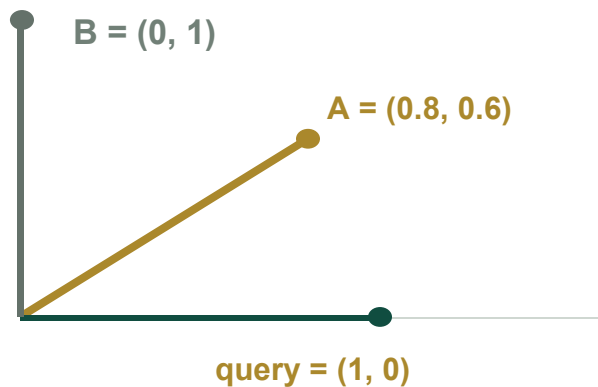
3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer



$$\cos(q, d) = (q \cdot d) / (|q| |d|)$$

A: 0.8

B: 0.0

A: project deadline B: reading list

A scores higher than B. Retrieve the original text behind A.

Retrieve the most similar text

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

COMP7607A

2026–27

Final project deadline

27 Nov, 17:00 HKT

0.8

**Final project
deadline**

0.0

Course reading list

`hits = index.search(encoder.encode(question), top_k=3)`

An answer with a source

QUERY **When is COMP7607A's final project due?**

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

COMP7607A

2026-27

Final project deadline

27 Nov, 17:00 HKT



27 Nov, 17:00 HKT

project_brief.pdf, p. 3

answer = llm(question, context=hits)

Relevant text can still miss the answer

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

COMP7607A

2026–27

Final project

Submit a report and code

No deadline here

Retrieve more evidence
or say it is missing.

Similar topics, different courses

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

LINGUISTICS

2026–27

Final project deadline

20 Nov, 17:00 HKT

COMP7607A / NLP

2026–27

Final project deadline

27 Nov, 17:00 HKT

Function names can differ by one letter

What does `get_user(id)` return?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

`get_user(id)`

One user profile

`get_users()`

A list of users

Keep identifiers intact during tokenisation.

Two searches, one candidate pool

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

BM25

COMP7607A

Embedding

“When must I hand it in?”

Reciprocal Rank Fusion (RRF) merges the two lists

Reciprocal Rank Fusion (RRF)

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

Rank	BM25	Embedding	RRF order
1	A / COMP7607A	B / Linguistics	A
2	D / Exam timetable	A / COMP7607A	B
3	B / Linguistics	C / Old copy	D

A reranker checks the full question

QUERY **When is COMP7607A's final project due?**

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

LINGUISTICS

2026–27

Final project deadline

20 Nov, 17:00 HKT

COMP7607A / NLP

2026–27

Final project deadline

27 Nov, 17:00 HKT

The right course moves to the top

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

COMP7607A / NLP

2026–27

Final project deadline

27 Nov, 17:00 HKT

LINGUISTICS

2026–27

Final project deadline

20 Nov, 17:00 HKT

1

The improved implementation

QUERY When is COMP7607A's final project due?

```
1 docs = clean_and_deduplicate(all_course_files)
2 chunks, index = build_index(docs)
3 q = rewrite(question, conversation)
4 candidates = rrf(bm25(q), dense(q))
5 evidence = rerank(q, candidates)[:k]
6 answer = llm(question, evidence)
```

Implementation challenges

Naive RAG

Three concrete failures. Three design decisions.

How do we choose chunk size and overlap?

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

COMP7607A

2026–27

Final project

27 Nov, 17:00 HKT

**Keep the date
with its heading.**

Why rerank only a short list?

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

Whole corpus

Retrieved candidates

10,000 documents

20 documents

Jointly score a small set. Missing evidence still needs retrieval.

Can more context make the answer worse?

QUERY When is COMP7607A's final project due?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

UNFILTERED CONTEXT

Old notices
Duplicate copies
Wrong courses

SELECTED EVIDENCE

Current course
Relevant section

More text can add noise and hide useful evidence.



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CONTEXT ENGINEERING · LECTURE 4

04

Graph-based RAG

SO FAR

We can retrieve individual course passages.

PART 4 OF 6



OUR NEXT QUESTION

**Which HKU courses can prepare me
for AI research?**

A learning path needs connections

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

**I want to become
an AI researcher.**

**Which courses?
In what order?
How are they connected?**

A graph makes relationships visible

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

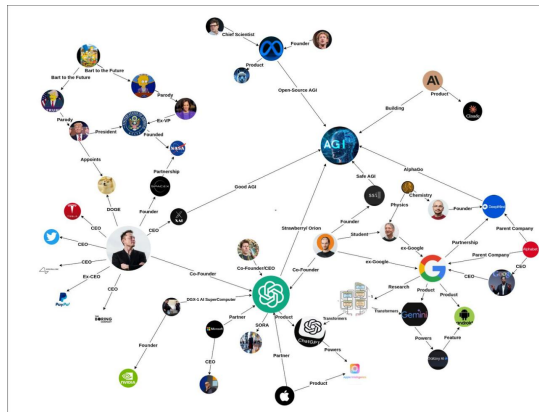
3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer



Nodes are entities. Edges describe their relationships.

LightRAG starts with the same course chunks

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

MATHS

2026–27

Linear algebra

Vectors and matrices

MACHINE LEARNING

2026–27

Learning models

Optimisation

NLP

2026–27

Language models

Builds on ML

Reuse the cleaned, deduplicated course corpus.

LightRAG: indexing, retrieval and generation

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

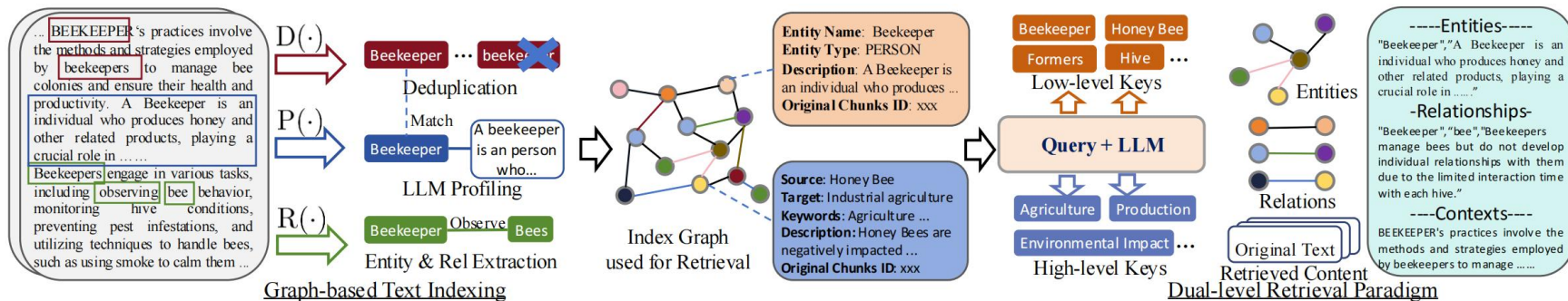
3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer



LightRAG extracts entities and relationships

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

NLP COURSE NOTES

Source B

NLP builds on

machine learning.

NLP

builds on

Machine learning

Inside LightRAG indexing

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

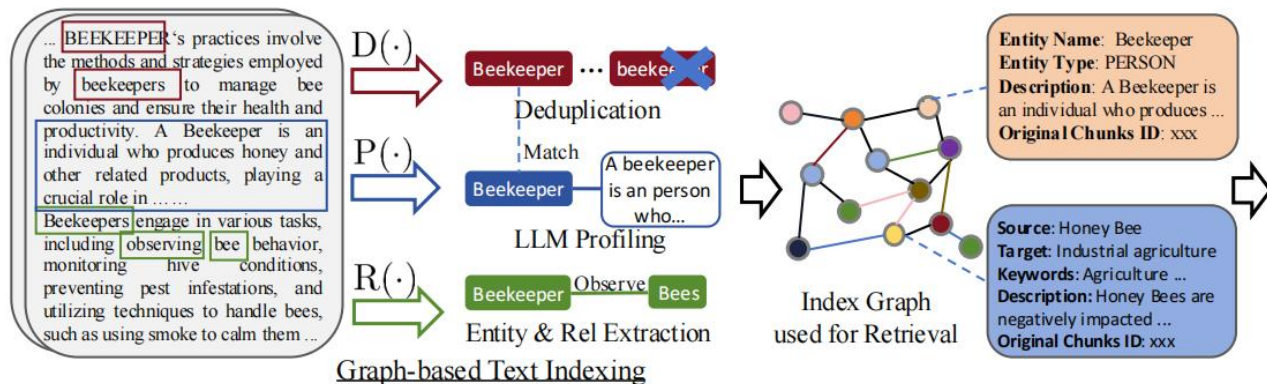
3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer



Extract relationships. Describe them. Merge duplicate entities.

What a real LightRAG graph looks like

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

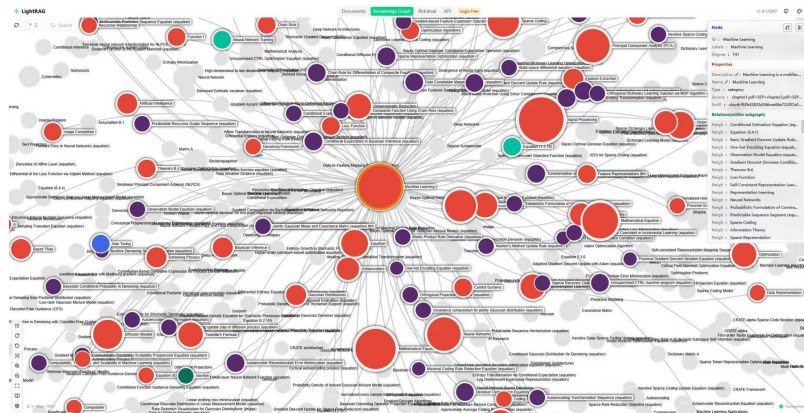
3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer



Each entity and relationship links back to supporting text.

A stable course ID prevents a wrong merge

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

SAME COURSE

COMP7607A
Natural Language
Processing

KEEP DISTINCT

Different course ID
Language Processing

Stable identifiers and source versions guide merging.

The same indexing step, a richer index

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

NAIVE RAG

**Text chunks
+ vectors**

LIGHTRAG

**Text chunks
+ entities and
relationships**

Both preserve source text and document metadata.

Graph storage keeps entities and relationships

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

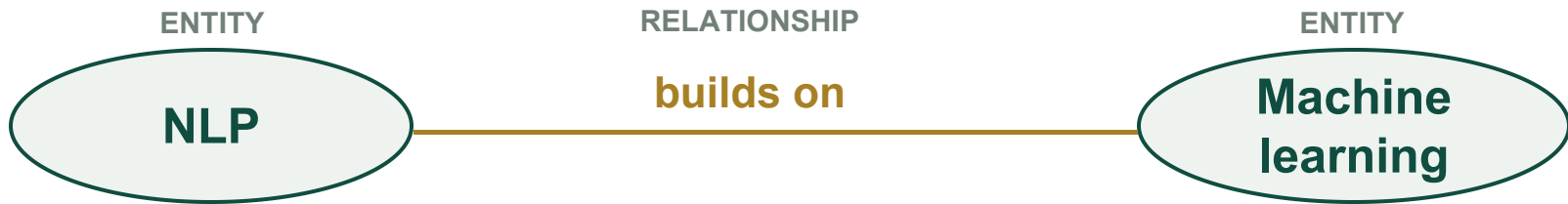
3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer



The relationship keeps its evidence

Description “NLP builds on machine learning.”

Source ID chunk_B · NLP course notes

Illustrative course relationship

Source: HKUDS / LightRAG

Graph and vector storage work together

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

STORAGE	WHAT IT CONTRIBUTES
Vector store	Find relevant entities and relationships by meaning
Graph store	Read their connections and source IDs
Text store (KV)	Fetch the original course chunks as evidence

Graph backend: NetworkX by default. Neo4j is optional.

LightRAG indexing in code

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

```
async def initialize_rag():
    rag = LightRAG(
        working_dir=WORKING_DIR,
        embedding_func=openai_embed,
        llm_model_func=gpt_4o_mini_complete,
    )
    # IMPORTANT: Both initialization calls are required!
    await rag.initialize_storages() # Initialize storage backends
    await initialize_pipeline_status() # Initialize processing pipeline
    return rag
```

await rag.ainsert(course_text)

await rag.finalize_storages()

Turn the question into two kinds of search clues

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

ENTITY KEYWORDS

Machine learning
NLP
Experiment design

RELATIONSHIP KEYWORDS

Research preparation
Learning progression

Both keyword levels use semantic retrieval in LightRAG.

LightRAG searches entities and relationships

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

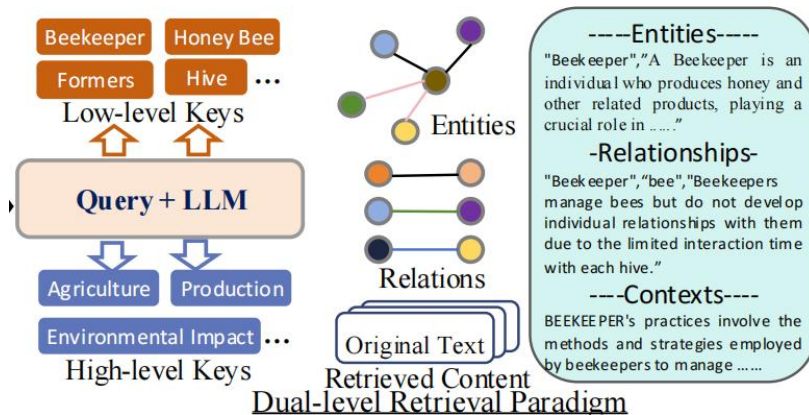
3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer



Specific concepts and broader relationships provide two search routes.

Relationship keywords find useful connections

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

**“research
preparation”**



**Relations about
foundations and
experiment design**

Connections still need source evidence

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

ML supports NLP

Source B

NLP COURSE NOTES

Source B

NLP builds on

machine learning.

The same retrieval step, connected evidence

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

NAIVE RAG

Similar text chunks

Select supporting text within a context budget.

LIGHTRAG

Relevant entities and relationships

LightRAG query modes in code

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

```
mode: Literal["local", "global", "hybrid", "naive", "mix", "bypass"] = "global"
    """Specifies the retrieval mode:
    - "local": Focuses on context-dependent information.
    - "global": Utilizes global knowledge.
    - "hybrid": Combines local and global retrieval methods.
    - "naive": Performs a basic search without advanced techniques.
    - "mix": Integrates knowledge graph and vector retrieval.
    """
```

Here, “hybrid” combines entity and relationship retrieval.

The answer explains a suggested learning order

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

Maths



ML



NLP



Research

Build foundations [A]

Learn modelling
[A,B]

Choose a topic [B]

Design experiments
[C]

Suggested path. Check your background and official prerequisites.

LightRAG vs Naive RAG

Table 1: Win rates (%) of baselines v.s. LightRAG across four datasets and four evaluation dimensions.

	Agriculture		CS		Legal		Mix	
	NaiveRAG	LightRAG	NaiveRAG	LightRAG	NaiveRAG	LightRAG	NaiveRAG	LightRAG
Comprehensiveness	32.4%	<u>67.6%</u>	38.4%	<u>61.6%</u>	16.4%	<u>83.6%</u>	38.8%	<u>61.2%</u>
Diversity	23.6%	<u>76.4%</u>	38.0%	<u>62.0%</u>	13.6%	<u>86.4%</u>	32.4%	<u>67.6%</u>
Empowerment	32.4%	<u>67.6%</u>	38.8%	<u>61.2%</u>	16.4%	<u>83.6%</u>	42.8%	<u>57.2%</u>
Overall	32.4%	<u>67.6%</u>	38.8%	<u>61.2%</u>	15.2%	<u>84.8%</u>	40.0%	<u>60.0%</u>

CS: 61.2%

Legal: 84.8%

LightRAG's overall preference win rate against Naive RAG.

LLM-judge preference, not factual accuracy.

Connections add indexing work

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

EXTRA INDEXING WORK

Extract relationships
Merge entities
Update affected records

POTENTIAL BENEFIT

Connected evidence
for broader questions

Graph construction costs more than a text-only vector index.

Implementation challenges

Graph-based RAG

Five concrete failures. Five design decisions.

When should we choose graph-based RAG?

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

SINGLE FACT

Final project deadline

CONNECTED QUESTION

**A path toward
AI research**

Measure the benefit against extraction and update costs.

What if the graph contains a wrong relation?

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

SOURCE

**This lecture covers
linear algebra.**

UNSUPPORTED CLAIM

**Students must complete
a specific maths course.**

Keep relation types distinct. Verify formal prerequisites against official course rules.

How do we limit graph retrieval noise?

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

TOO MUCH

Every neighbouring course
Every linked document

USEFUL CONTEXT

Goal-relevant relations
Supporting source chunks

Limit candidates and the context budget. Test whether the necessary evidence remains.

How do we update the graph?

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

OLD REQUIREMENT

NLP builds on topic A
2025 course material

UPDATED REQUIREMENT

NLP builds on topic B
2026 course material

Version the sources. Update affected entities, relations and links to the original text.

How do we verify a multi-hop answer?

Which HKU courses can prepare me for AI research?

1 Collect

2 Clean

3 Chunk

4 Index

5 Retrieve

6 Rank

7 Answer

GRAPH CONNECTION

Maths links to ML
ML links to NLP

CHECK THE ADVICE

Evidence for each step
Student background and rules

Check source support and prerequisites. Present the result as a recommendation.

Could a smaller local model do the job?

Smaller model
Less memory
Private local data

**Can it still find
the right evidence?**

MiniRAG keeps the original text close to the entities

Course entity



Original chunk A
Original chunk B

github.com/HKUDS/LightRAG
github.com/HKUDS/MiniRAG



香港大學
THE UNIVERSITY OF HONG KONG

CONTEXT ENGINEERING · LECTURE 4

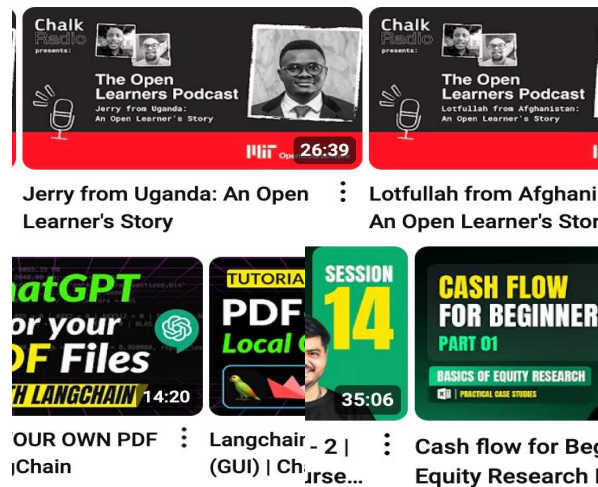
05

Multimodal RAG

SO FAR

Our evidence has come from course documents.

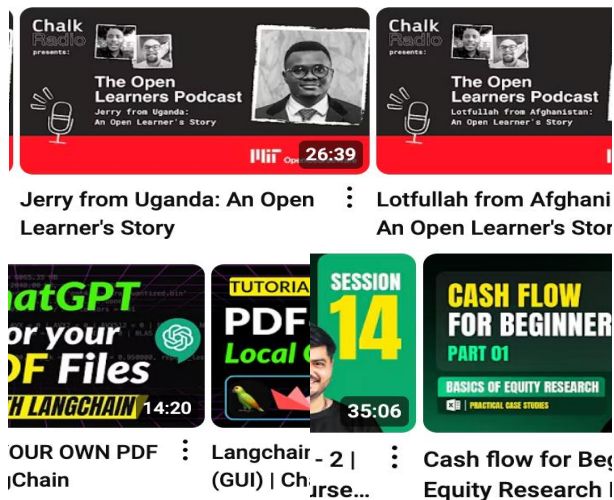
PART 5 OF 6



OUR NEXT QUESTION

How can we find a useful moment inside many videos?

Useful explanations also live in videos



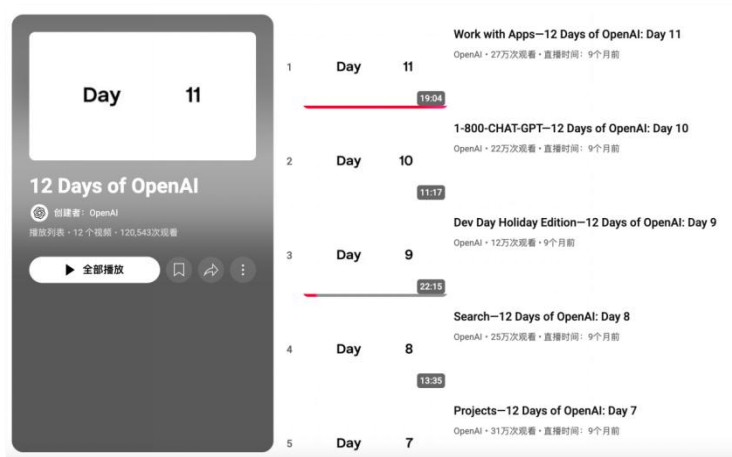
A useful explanation
may be spoken, drawn
or demonstrated.

One question across a collection of tutorials

How are model answers scored in reinforcement fine-tuning?

Input: 12 Days of OpenAI
A collection of tutorials

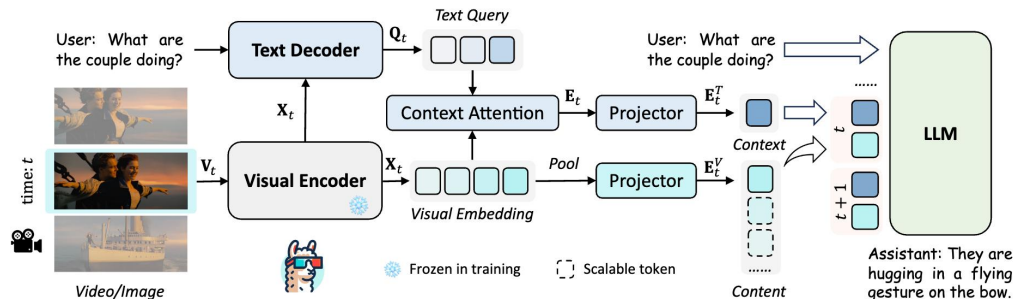
Find the explanation
and return its location.



A whole video collection is expensive to inspect at once

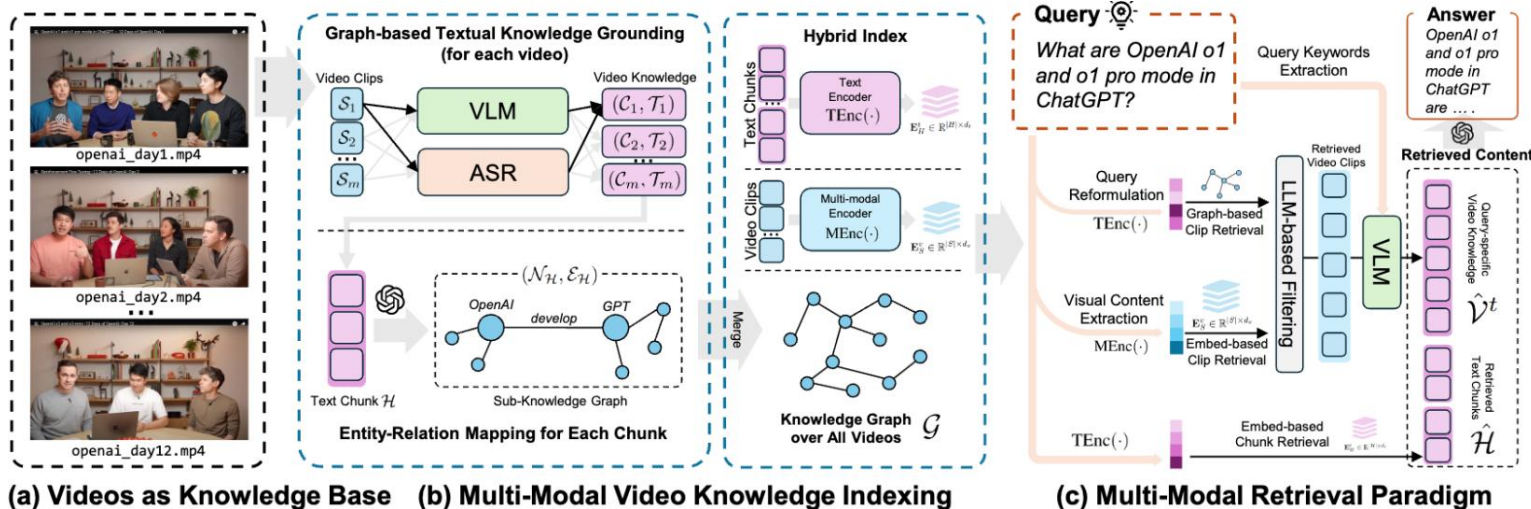
How are model answers scored in reinforcement fine-tuning?

More videos
More frames
More computation



VideoRAG: three familiar modules

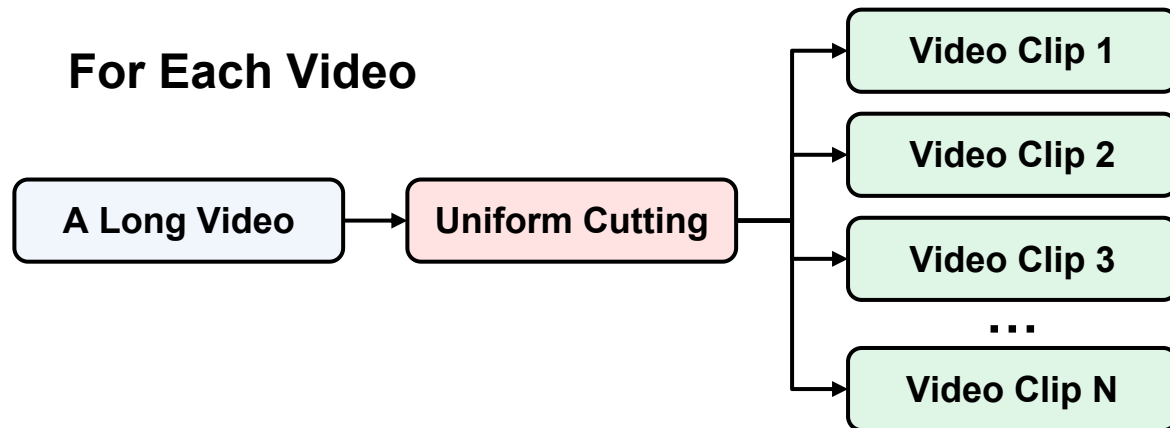
How are model answers scored in reinforcement fine-tuning?



Indexing starts with time-stamped clips

How are model answers scored in reinforcement fine-tuning?

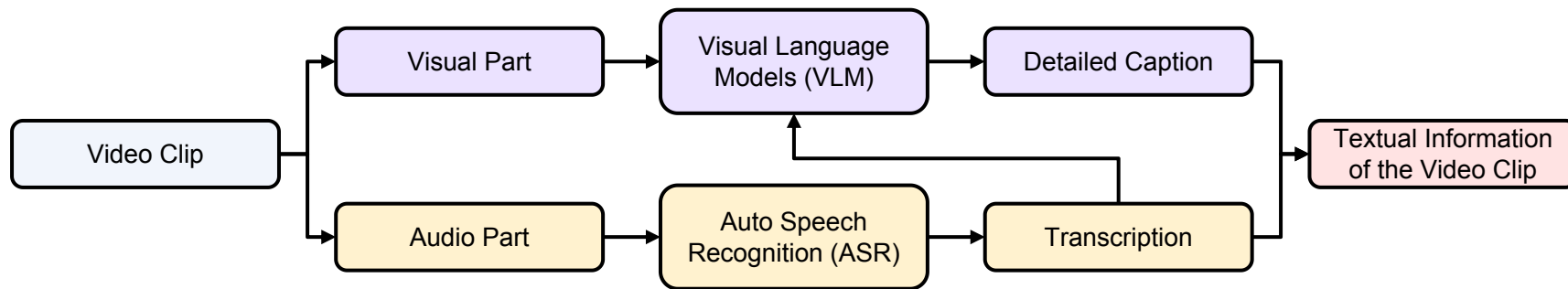
INDEX / SEGMENT



Each clip has speech and visual information

How are model answers scored in reinforcement fine-tuning?

INDEX / ASR + VLM



Hear the explanation. See what it refers to.

How are model answers scored in reinforcement fine-tuning?

INDEX / SPEECH + FRAMES



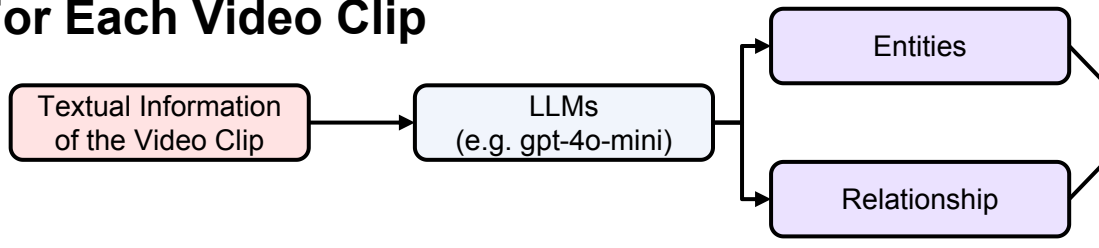
Audio tells us what is said. Frames show what is demonstrated.

Shared concepts connect clips across videos

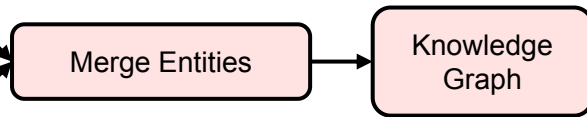
How are model answers scored in reinforcement fine-tuning?

INDEX / GRAPH

For Each Video Clip



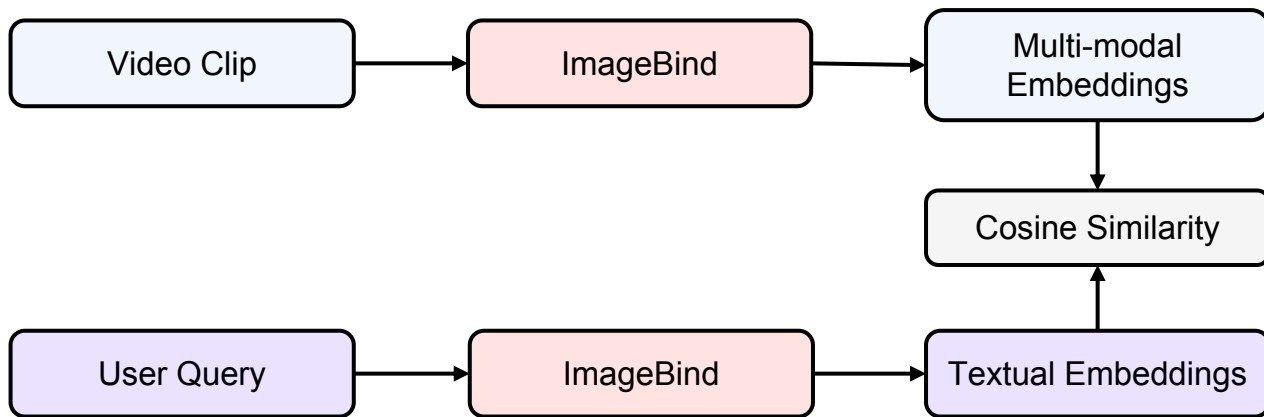
For all Video Clips



Visual retrieval compares queries with video clips

How are model answers scored in reinforcement fine-tuning?

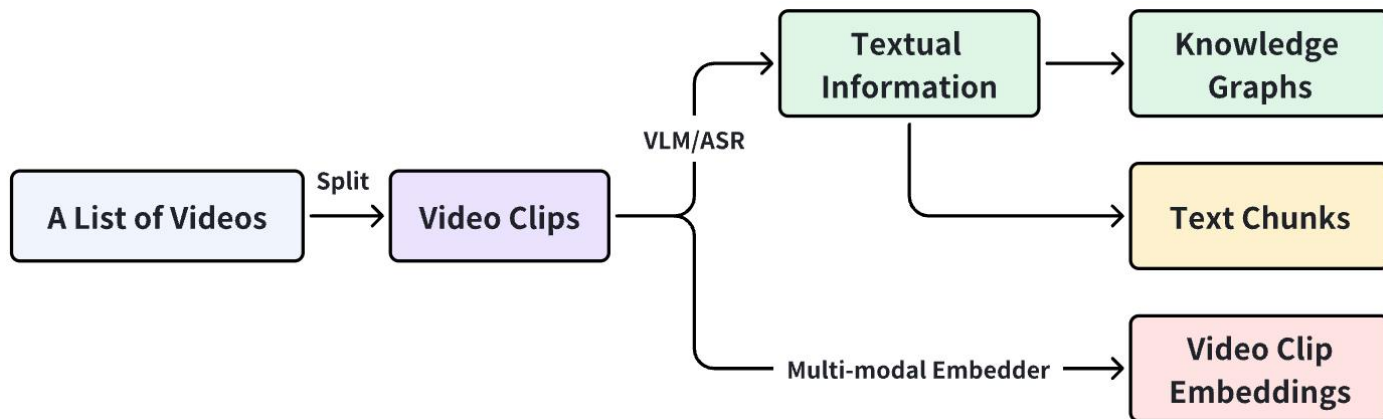
INDEX / MULTIMODAL EMBEDDING



The video index keeps several views of the same clip

How are model answers scored in reinforcement fine-tuning?

INDEX / SUMMARY

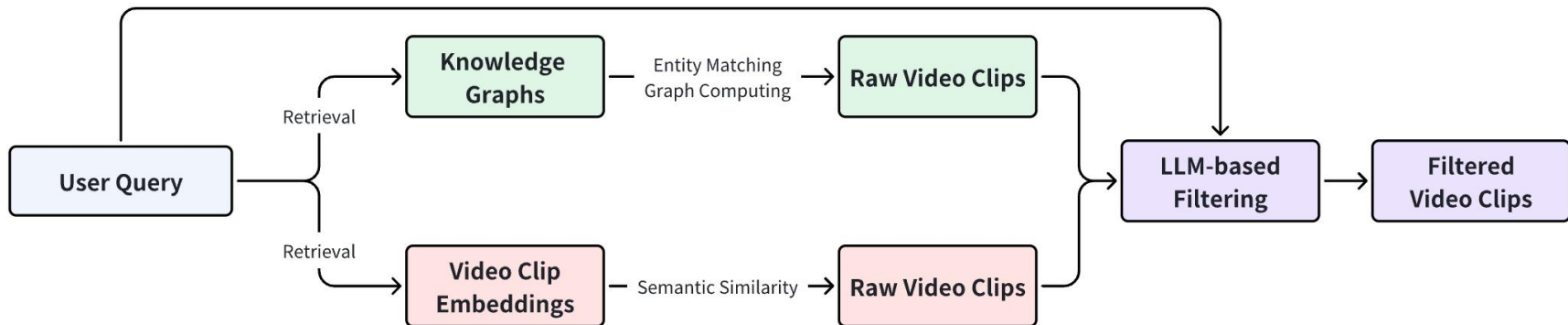


Two retrieval routes find candidate clips

How are model answers scored in reinforcement fine-tuning?

RETRIEVE

Video Clips Retrieval

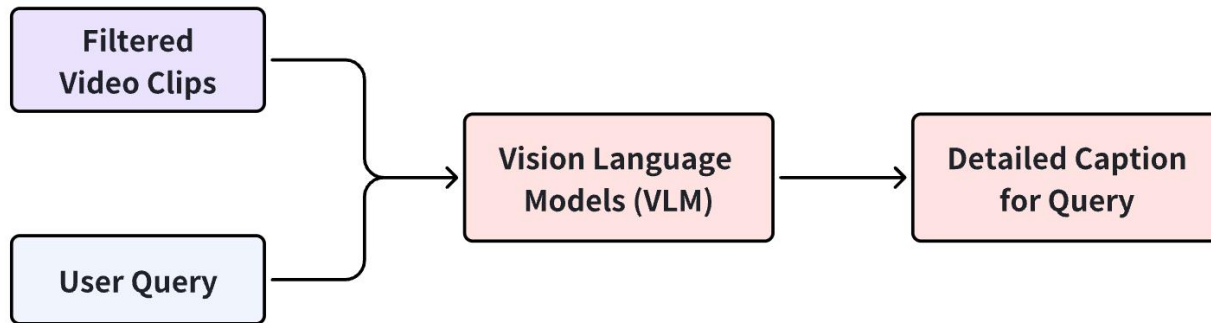


The model revisits selected clips with the question

How are model answers scored in reinforcement fine-tuning?

RETRIEVE / INSPECT

Detailed Caption Generation

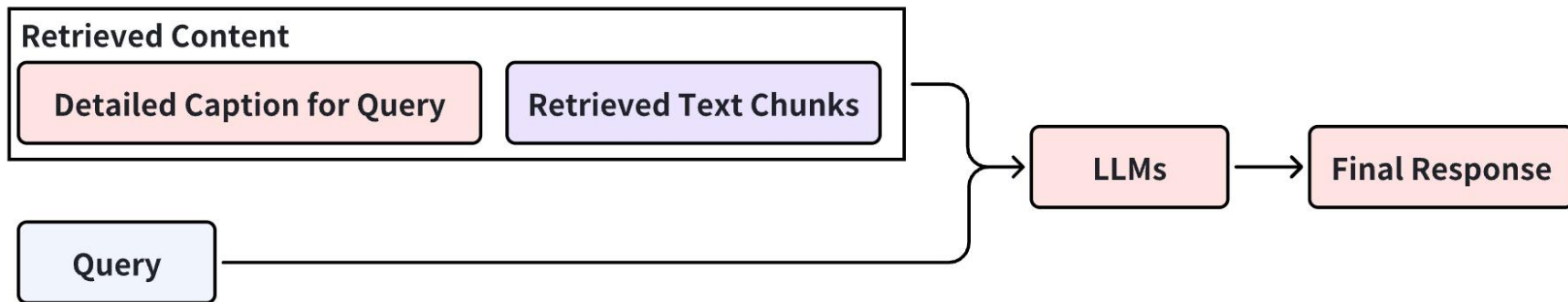


Generation uses detailed captions and retrieved text

How are model answers scored in reinforcement fine-tuning?

GENERATE

Retrieval-Augmented Generation (RAG) with Videos

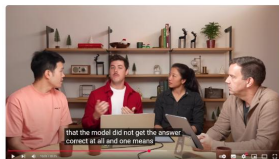


The answer should take us back to the video

How are model answers scored in reinforcement fine-tuning?

GENERATE / CITE

A grader scores the model's response.



OpenAI Day 2 / 10:00–12:00
Reported retrieval example

Implementation challenges

Multimodal RAG

Five questions about text, images and documents.

When do we need multimodal RAG?

MULTIMODAL RAG / CHALLENGE 1

WORDS CONTAIN THE ANSWER

“Submit by 27 November.”

Text retrieval can be enough.

Use visual evidence when words alone leave out the answer.

THE PICTURE CONTAINS THE ANSWER

“Which layer connects to the classifier?”

Retrieve and inspect the diagram.

Can OCR alone handle a visual PDF?

MULTIMODAL RAG / CHALLENGE 2

FLATTENED OCR

0.92 0.86
12 8

	Model A	Model B
Accuracy	0.92	0.86
Time (s)	12	8

Which value belongs to Model A?

Keep headers and layout. Verify the original page.

Image embeddings or parsed text?

MULTIMODAL RAG / CHALLENGE 3

PARSE FIRST

Text + structured tables

Inspect extracted fields.
Check parsing errors.

EMBED THE PAGE IMAGE

Layout + visual content

Keep the page to inspect.
Measure storage and latency.

Compare both on the same questions. A hybrid can help.

How should we chunk a mixed document?

MULTIMODAL RAG / CHALLENGE 4

A CHART NEEDS ITS CONTEXT

Evidence unit	Keep together
Table	Headers + rows + units
Figure	Image + caption + nearby text
Source	Document ID + page number

Split by content boundaries. Preserve links to the original page.

How do we evaluate multimodal RAG?

MULTIMODAL RAG / CHALLENGE 5

QUESTION: WHICH MODEL HAS HIGHER ACCURACY?

Check	What we verify
Retrieval	Did we retrieve the correct table?
Correctness	Does the answer say Model A?
Faithfulness	Does the cited table support it?

Use labelled questions and evidence. Audit with human checks.

RAG Evaluation

Did we find the evidence?

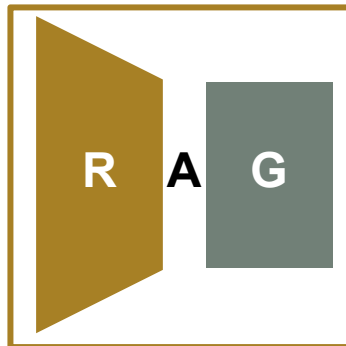
Did we use it correctly?

Retrieval quality, answer quality and system efficiency.

RAG evaluation asks three different questions

Can we find the right
information

Can we generate high-quality
answers?



Is the system efficient and
practical?

Did retrieval find enough useful evidence?

QUERY When is COMP7607A's final project due?

EVALUATION / PRECISION + RECALL

Labelled relevant passages	Retrieved top 3
S1, S2, S3, S4	S1, S3, X

Precision@3 = 2 / 3

Recall@3 = 2 / 4

Useful evidence should appear near the top

QUERY When is COMP7607A's final project due?

EVALUATION / MRR + NDCG

Rank	Returned evidence	Relevance grade
1	Irrelevant notice	0
2	Exact current deadline	2
3	Related project detail	1

RR = 1 / 2 NDCG@3 $\approx$ 0.66

MRR averages RR across questions. NDCG rewards useful evidence near the top.

A multi-step answer needs every necessary source

Which HKU courses can prepare me for AI research?

EVALUATION / ALL-RECALL

Retrieved: A + B Still missing: C



Maths supports ML

ML supports NLP

Experiment design

All-Recall = 0 for this question

The same date can be written in different ways

QUERY When is COMP7607A's final project due?

EVALUATION / GENERATION

REFERENCE

27 November, 17:00 HKT

MODEL ANSWER

**5 pm Hong Kong time
on November 27**

Normalize the date and time before checking factual agreement.

Exact Match: identical text Token F1 / ROUGE-L: wording overlap

Can a faithful answer still be wrong?

QUERY When is COMP7607A's final project due?

EVALUATION / ANSWER QUALITY

ANSWER + OLD SOURCE

“20 November.”
Matches last year's file.

CURRENT GROUND TRUTH

27 November
Current course version

Check factual correctness, source support and source freshness separately.

How can we trust an LLM judge?

QUERY When is COMP7607A's final project due?

EVALUATION / HUMAN + LLM JUDGE

EVALUATE THE ANSWER

Coverage and correctness
Helpful reasoning
Evidence support

CHECK THE JUDGE

Blind A/B order
Consistent rubric
Human spot checks

A preference win rate measures the judge's choice, not verified factual accuracy.

How do we measure speed and cost?

QUERY When is COMP7607A's final project due?

EVALUATION / SYSTEM EFFICIENCY

Measure	Keep separate
Indexing cost	Parsing, embeddings, graph extraction
Query latency	Retrieval, reranking, generation
Per-question cost	Input/output tokens + retrieval services

How do we build a RAG test set?

EVALUATION / PRACTICE

Question type	Ground truth to prepare
Final project deadline	Current date + source passage
AI research learning path	Required evidence + constraints
Video explanation	Relevant clip + supported answer

The question determines which evidence we need

FIND A FACT

**Final project
deadline**

CONNECT THE FACTS

**A path toward
AI research**

FIND A VIDEO MOMENT

**A grader
explained on
screen**

Find the evidence. Check the answer and its source.